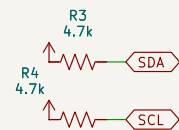
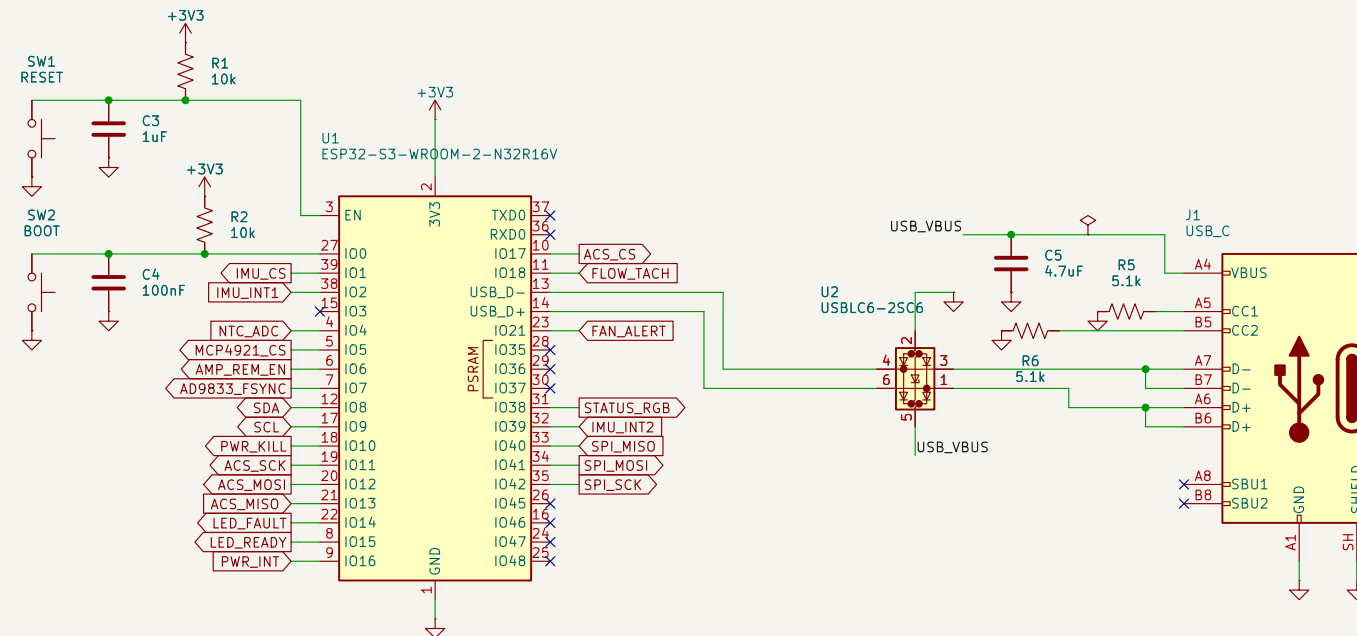


GPIO map follows include/config/pin\_config.h

A circuit diagram showing a parallel combination of two capacitors, C1 and C2. C1 is labeled '10uF' and C2 is labeled '100nF'. Both capacitors are connected between a +3V3 supply line and a common ground line. The ground is indicated by a triangle symbol.



File: sheets/power\_input.kicad\_sch

File: sheets/power\_button.kicad\_sch

File: sheets/rails\_logic.kicad\_sch

File: sheets/rails\_15v.kicad\_sch

File: sheets/signal\_gen.kicad\_sch

File: sheets/amp\_control.kicad\_sch

File: sheets/amp\_current.kicad\_sch

File: sheets/system\_current.kicad\_sch

File: sheets/cooling.kicad\_sch

File: sheets/cold\_head.kicad\_sch

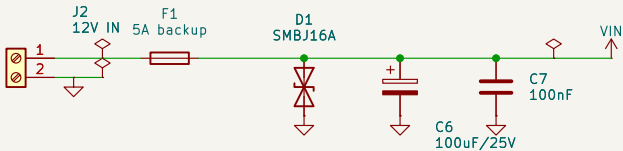
File: sheets/imu.kicad\_sch

File: sheets/indicators.kicad\_sch

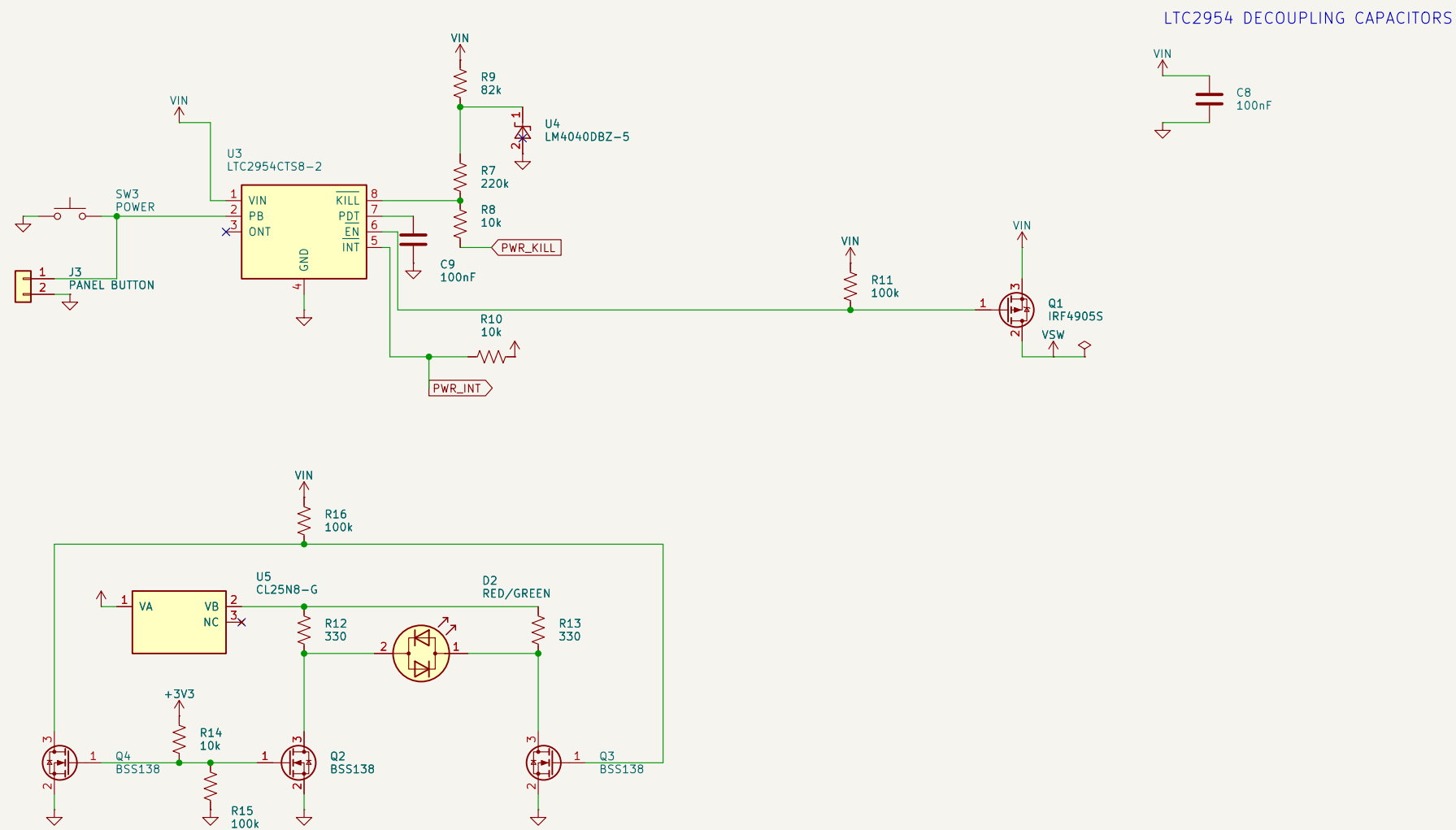
Rev: A  
Id: 1/13

12 V input from a Dell XYK93 (DPS–2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS\_ON held low.

F1 is a backup only: add an inline high–breaking–capacity fuse (7.5–10 A) at the PSU end. Reverse polarity: TVS forward–clamps and blows the fuse.



PWR\_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby

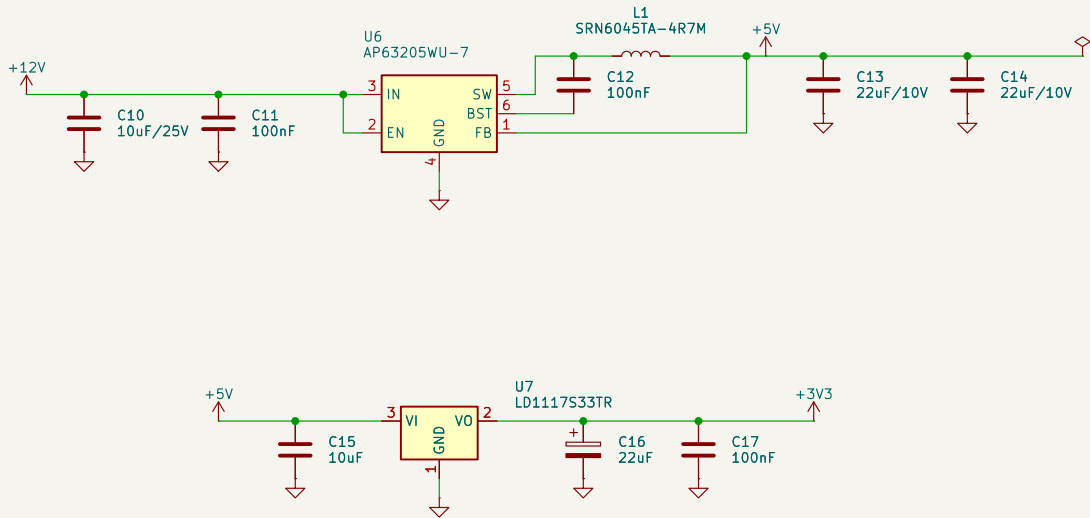


Rev: A

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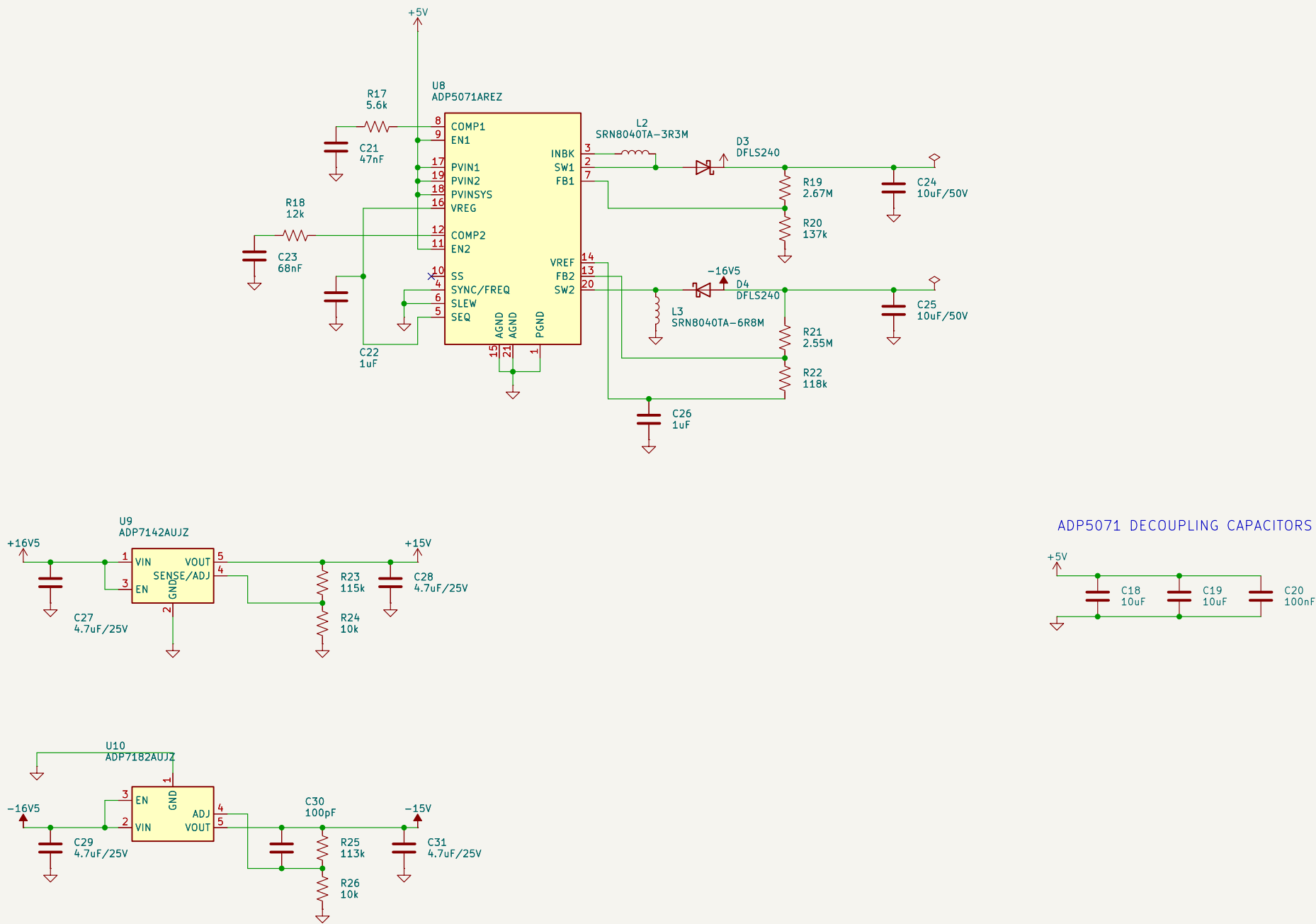
12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.



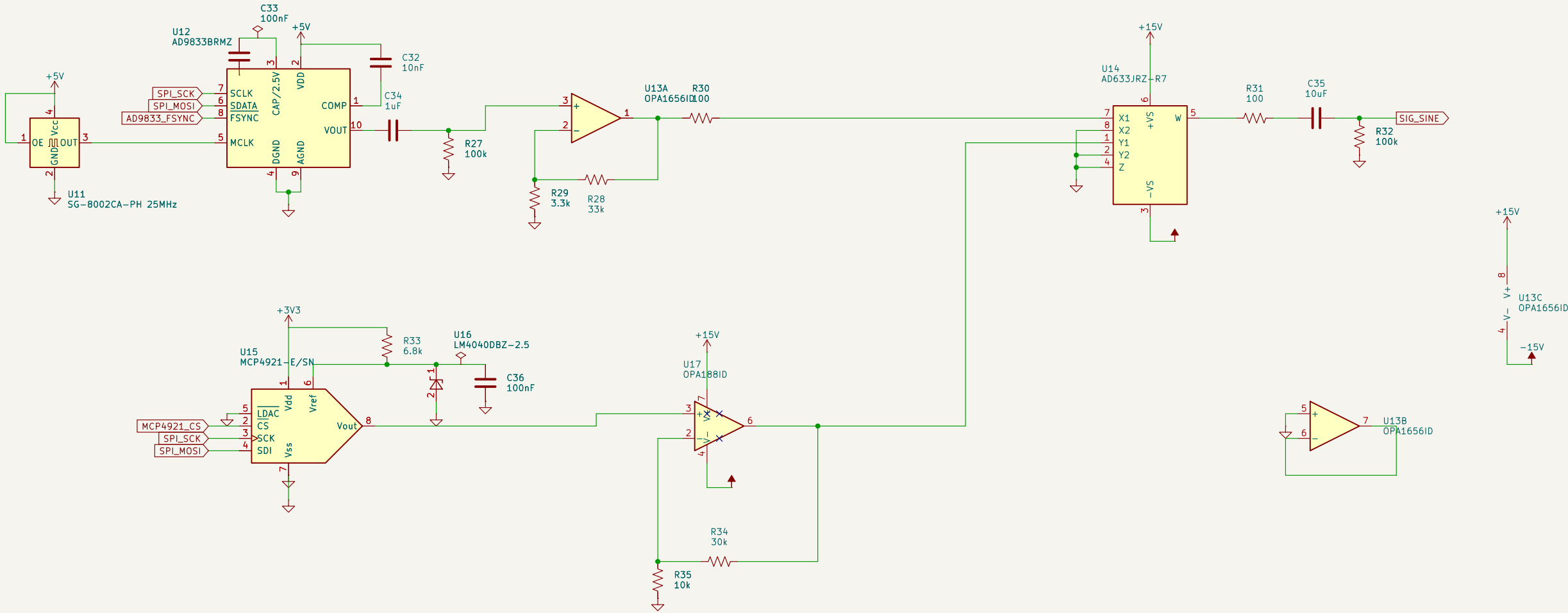
ADP5071AREZ (5 V in): +16.5 V boost and −16.5 V inverting rails; ADP7142 (+15 V) and ADP7182 (−15 V) post-regulate.

Values from ADP5071 datasheet Fig. 47 / Tables 9–11 (+/−16.5 V output rows rescaled via RFT). ADP7182 EN tied to VIN (auto-start); |VEN| >= 2 V enables.

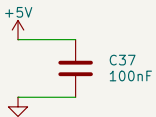


AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0-10 V amplitude control.

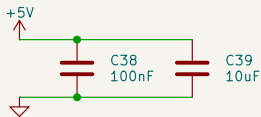
Vctl 0-10 V (2.5 V ref x4) x sine +/-3.3 V pk / 10 = 0..3.3 V pk to amplifier input



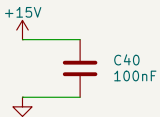
SG-8002CA DECOUPLING CAPACITORS



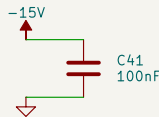
AD9833 DECOUPLING CAPACITORS



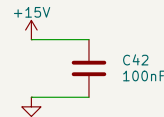
OPA1656 +15V DECOUPLING CAPACITORS



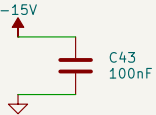
OPA1656 -15V DECOUPLING CAPACITORS



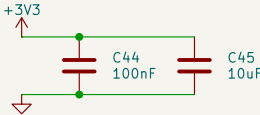
AD633 +15V DECOUPLING CAPACITORS



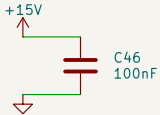
AD633 -15V DECOUPLING CAPACITORS



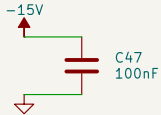
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 -15V DECOUPLING CAPACITORS



**Cryocooler Controller**

Sheet: /Signal Generator/  
File: signal\_gen.kicad\_sch

**Title: Signal Generator**

Size: A3

Date:

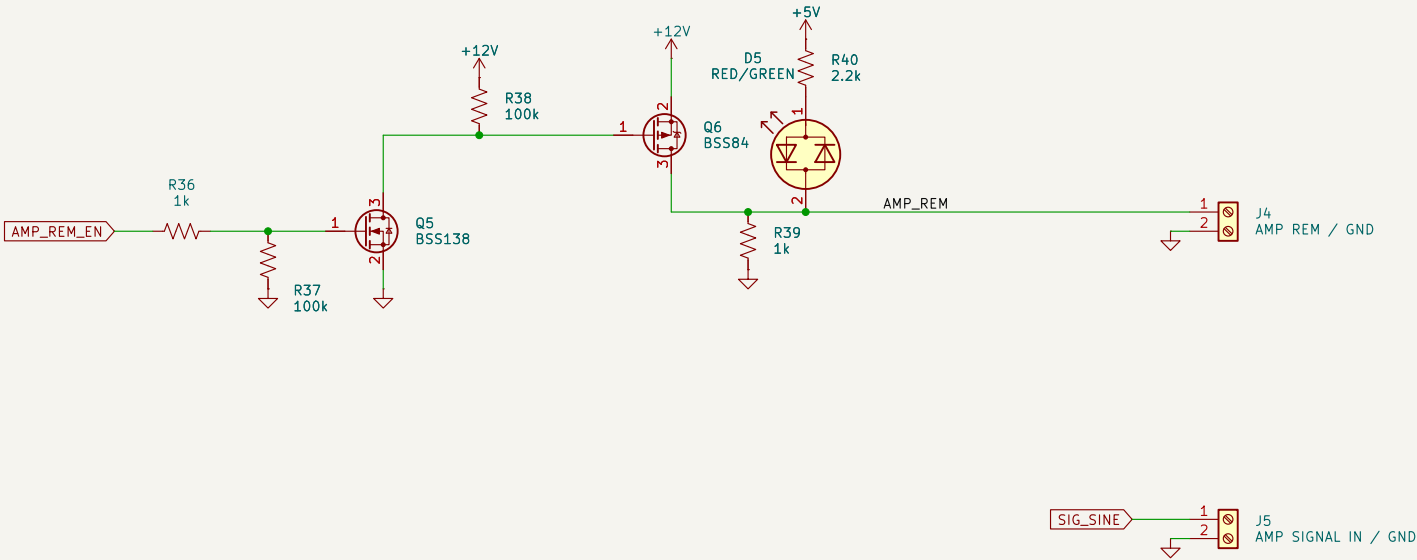
Rev: A

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Id: 6/13

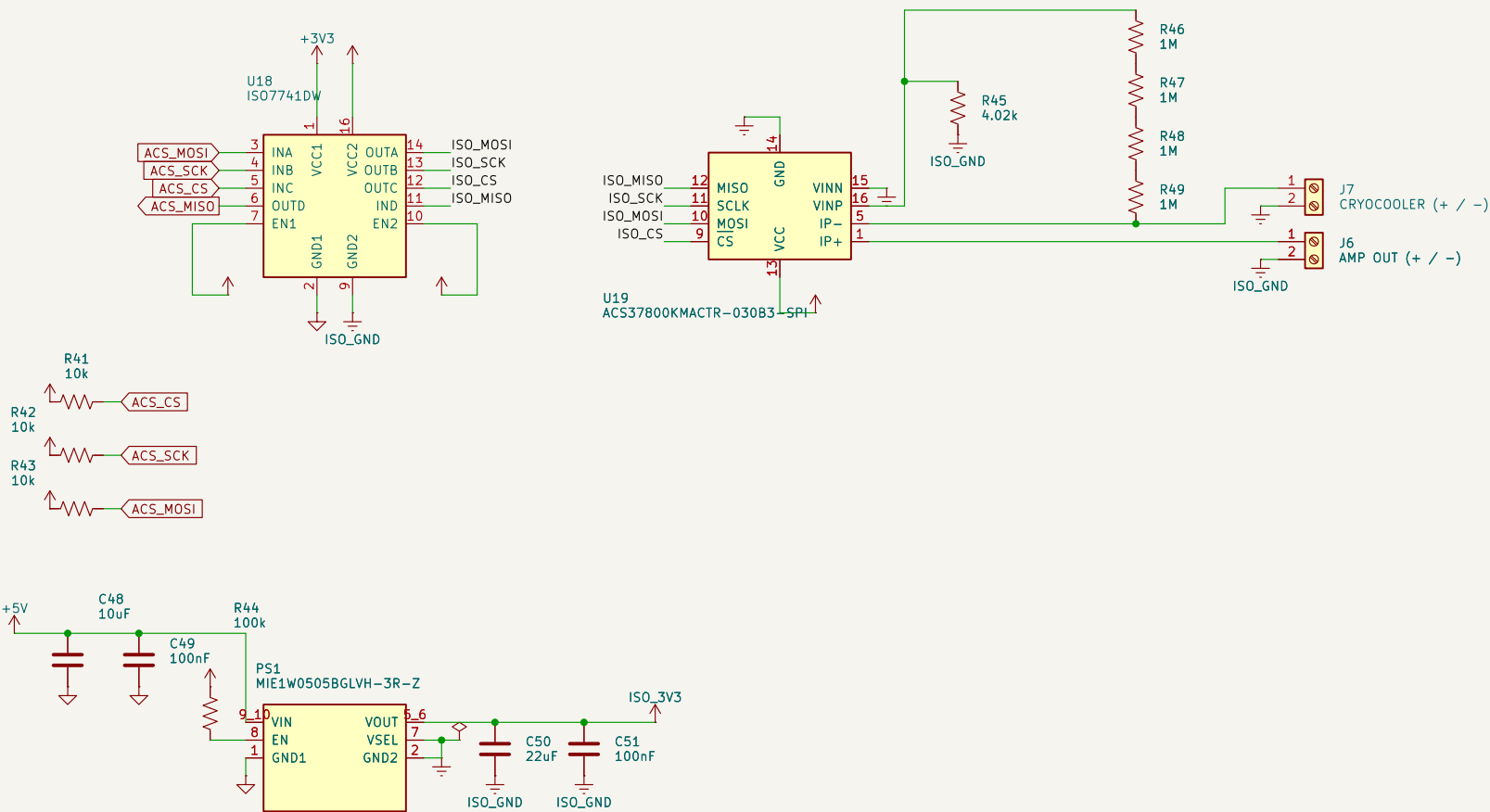
GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.

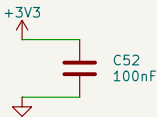


Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

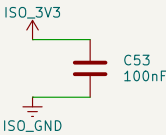
Mains-class creepage/clearance applies between ISO\_GND domain and the logic side on the PCB.



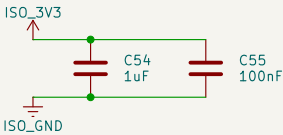
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS

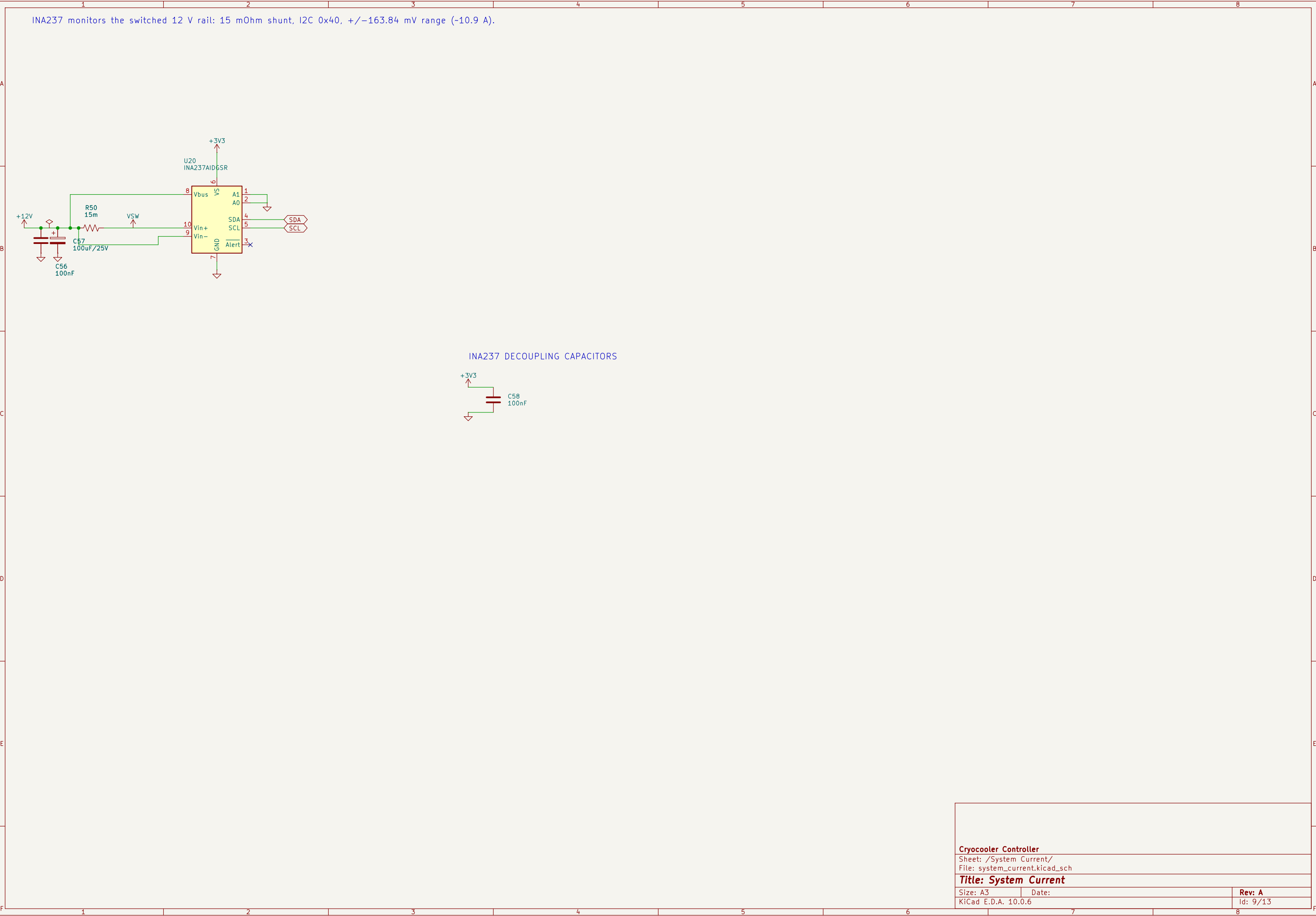


Cryocooler Controller

Sheet: /Amp Current/  
File: amp\_current.kicad\_sch

Title: Amp Current

|                     |       |          |
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| Size: A3            | Date: | Rev: A   |
| KiCad E.D.A. 10.0.6 |       | Id: 8/13 |



EMC2303: ADDR\_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.

The schematic diagram illustrates the Cryocooler Controller circuit, centered around the EMC2303 (U21) microcontroller. The microcontroller is configured with the following components:

- Power and Ground:** VDD is connected to +3V3, and GND is connected to ground.
- Communication:** SMDATA (pin 1) is connected to SDA (pin 1) via a 33k resistor (R51). SMCLK (pin 2) is connected to SCL (pin 2) via a 33k resistor (R51).
- Control Signals:** ADDR\_SEL (pin 4) is connected to +3V3 via a 4.7k resistor (R52). CLK (pin 9) is connected to +3V3 via a 10k resistor (R53). ALERT (pin 10) is connected to a fan fault indicator (D6) via a 1k resistor (R54).
- Fan and Pump Control:** PWM1 TACH1 (pin 5) is connected to +3V3 via a 10k resistor (R55). PWM2 TACH2 (pin 7) is connected to +3V3 via a 4.7k resistor (R56). PWM3 TACH3 (pin 11) is connected to +3V3 via a 10k resistor (R57). PWM4 TACH4 (pin 12) is connected to +3V3 via a 4.7k resistor (R58).
- Decoupling Capacitors:** C59 (1uF) and C60 (100nF) are connected to +3V3 and ground. C61 (22uF/25V) and C62 (22uF/25V) are connected to +12V and ground. C63 (100nF) is connected to +3V3 and ground.
- Connectors:** J8 (FAN 4-pin PWM) and J9 (PUMP 4-pin PWM) are connected to the microcontroller's PWM outputs. J10 (ALPHACOOL ES FLOW 3-pin) and J11 (ALPHACOOL ES NTC 10k) are connected to the microcontroller's TACH inputs.

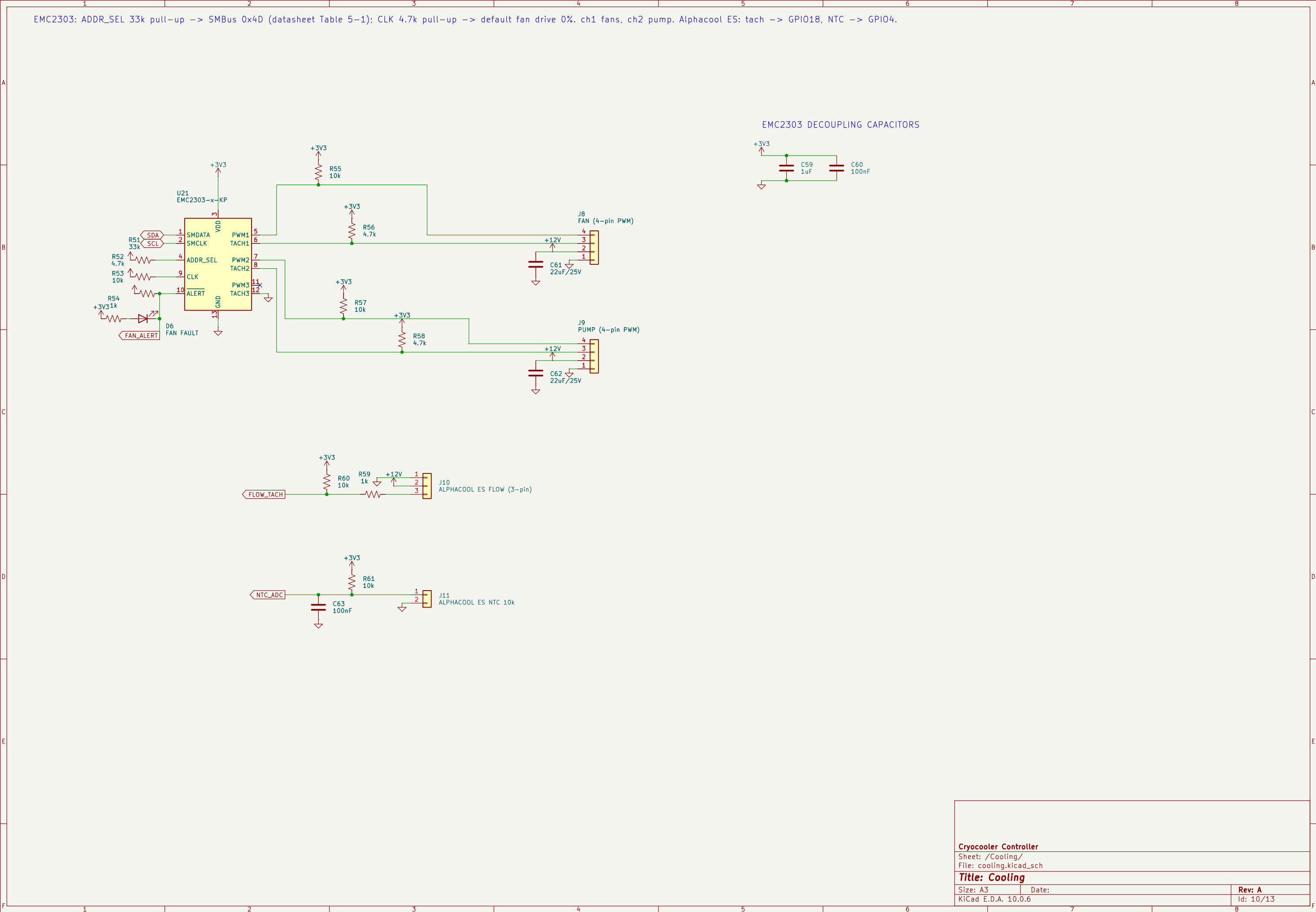
**EMC2303 DECOUPLING CAPACITORS**

The diagram shows the decoupling capacitors for the EMC2303 microcontroller. C59 (1uF) and C60 (100nF) are connected to +3V3 and ground. C61 (22uF/25V) and C62 (22uF/25V) are connected to +12V and ground. C63 (100nF) is connected to +3V3 and ground.

**Legend:**

- U21: EMC2303-x-KP
- R51: 33k
- R52: 4.7k
- R53: 10k
- R54: 1k
- R55: 10k
- R56: 4.7k
- R57: 10k
- R58: 4.7k
- C59: 1uF
- C60: 100nF
- C61: 22uF/25V
- C62: 22uF/25V
- C63: 100nF
- D6: FAN FAULT
- J8: FAN (4-pin PWM)
- J9: PUMP (4-pin PWM)
- J10: ALPHACOOL ES FLOW (3-pin)
- J11: ALPHACOOL ES NTC 10k

**Sheet: /Cooling/**  
**File: cooling.kicad\_sch**  
**Title: Cooling**  
Size: A3 Date: Rev: A  
KiCad E.D.A. 10.0.6 Id: 10/13



Sheet: /Cooling/  
File: cooling.kicad\_sch

**Title: Cooling**

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| Size: A3            | Date: | Rev: A    |
| KiCad E.D.A. 10.0.6 |       | Id: 10/13 |

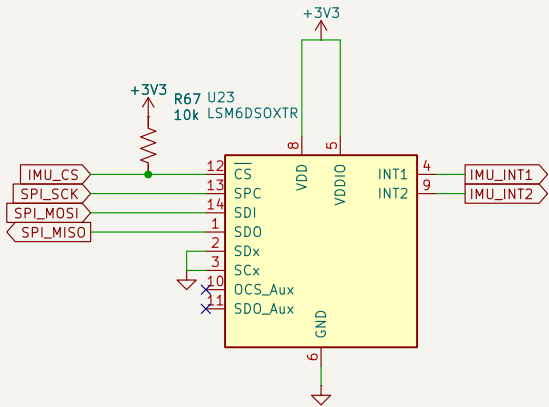
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File: cooling.kicad\_sch

**Title: Cooling**

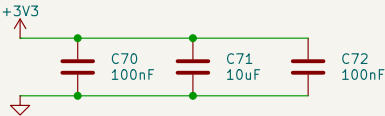
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| Size: A3            | Date: | Rev: A    |
| KiCad E.D.A. 10.0.6 |       | Id: 10/13 |

[illegible]

LSM6DSOX in 4-wire SPI mode on the main bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).



LSM6DSOX DECOUPLING CAPACITORS

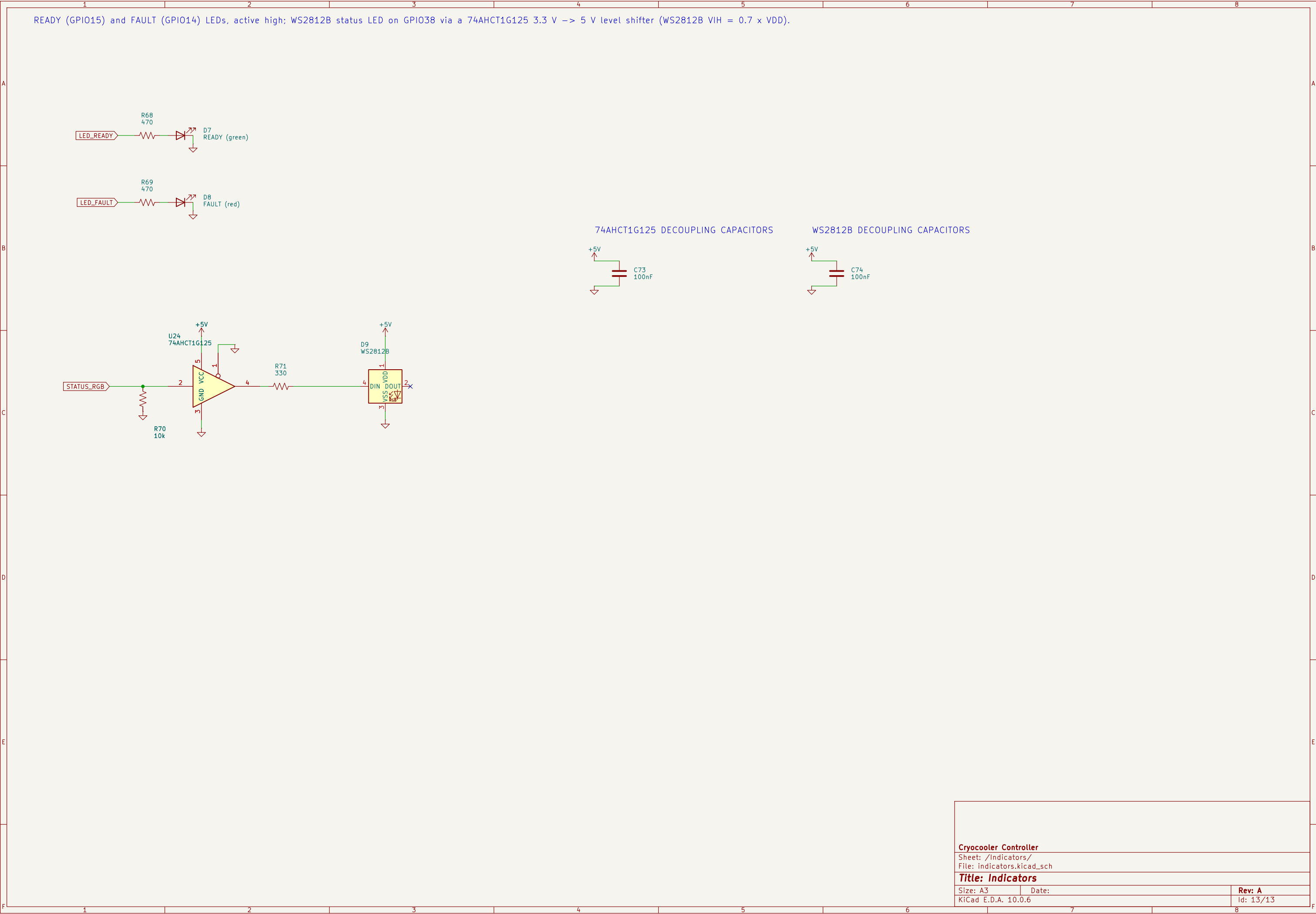


Cryocooler Controller

Sheet: /IMU/  
File: imu.kicad\_sch

Title: IMU

|                     |       |           |
|---------------------|-------|-----------|
| Size: A3            | Date: | Rev: A    |
| KiCad E.D.A. 10.0.6 |       | Id: 12/13 |



**Cryocooler Controller**

Sheet: /Indicators/  
File: indicators.kicad\_sch

**Title: Indicators**

|                     |       |           |
|---------------------|-------|-----------|
| Size: A3            | Date: | Rev: A    |
| KiCad E.D.A. 10.0.6 |       | Id: 13/13 |